Tampered Image Detection And Localization - A Systematic Ablation Study Using Deep Learning

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***Abstract*—Digital images can be manipulated in ways that are invisible to the human eye. Detecting such tampering and pinpointing exactly which pixels were altered is a challenging problem in image forensics. This report presents a systematic ablation study for pixel-level image tamper detection and localization on the CASIA v2.0 benchmark dataset (12,614 images). The final system uses a UNet segmentation model with a pretrained ResNet-34 encoder, Error Level Analysis (ELA) preprocessing, and CBAM attention modules. Through 60+ controlled experiments across four research tracks, we established a clear finding: input representation matters most. Switching from raw RGB images to Multi-Quality RGB ELA (9-channel) input produced a +34.19 percentage point improvement in Pixel F1 — more than all other improvements combined. The best model (vR.P.19) achieves a Pixel F1 of 0.7965, IoU of 0.6615, and Pixel AUC of 0.9665. Every experiment followed a strict single-variable ablation discipline (one change per experiment), enabling clear cause-and-effect analysis.**

***Index Terms*—Image tampering detection, image forgery localization, error level analysis, UNet segmentation, ablation study, CASIA dataset.**

I. Introduction

Digital images are ubiquitous in journalism, law enforcement, scientific research, and social media. However, modern image editing tools make it easy to manipulate photographs in ways that are nearly undetectable by the human eye. Common types of tampering include:

- **Splicing:** Copying a region from one image and pasting it into another.
- **Copy-move:** Duplicating a part of an image within the same image, often to hide or fabricate content.

Detecting whether an image has been tampered with (*detection*) is useful, but knowing *where* the tampering occurred (*localization*) is far more valuable. This project tackles both tasks: given an input image, the system produces a binary mask highlighting the tampered pixels.

Building such a system is difficult for several reasons. First, tampered regions often occupy a very small fraction of the image (sometimes less than 2% of all pixels), creating a severe class imbalance at the pixel level. Second, skilled forgers adjust lighting, color, and compression to make edits blend seamlessly with the surrounding content. Third, JPEG compression artifacts can either mask or mimic tampering traces, complicating forensic analysis.

This report describes a systematic approach to solving this problem. Rather than building a single model and reporting its results, we conducted **60+ controlled experiments** across four research tracks, changing exactly one variable at a time. This ablation methodology allows us to determine *what actually works and why*, rather than relying on guesswork or combining multiple changes simultaneously.

The key contributions of this work are:

- A complete pipeline from raw images to pixel-level tamper localization using ELA preprocessing and pretrained UNet segmentation.
- A rigorous ablation study establishing that **input representation dominates all other factors** — the single biggest improvement came from how we preprocess the image, not from architecture or training tricks.
- Transparent documentation of both successes and failures, including three failed research tracks that provided lessons directly shaping the final approach.
- Full reproducibility via Weights & Biases experiment tracking and fixed random seeds.
- Open-source code and experiment logs available publicly.[1,2]

II. Related Work

Image forgery detection has been studied extensively over the past decade. This section reviews the main approaches that influenced our work.

A. Traditional Forensic Methods

Early forgery detection relied on hand-crafted features such as JPEG compression artifacts [8], noise inconsistencies, and Color Filter Array (CFA) patterns. Error Level Analysis (ELA) [2] is one such technique: it re-saves an image at a given JPEG quality and measures the difference from the original. Tampered regions that were compressed at different quality levels show inconsistent residuals. While these methods are interpretable, they struggle with sophisticated forgeries and produce high false-positive rates when used alone.

B. CNN-Based Approaches

Deep learning transformed forgery detection by learning features directly from data. Nagm et al. [3] combined ELA preprocessing with a custom CNN for binary classification

[1] GitHub: https://github.com/The-Harsh-Vardhan/TIDAL-Tampered_Image_Detection_And_Localization

[2] W&B Report: https://api.wandb.ai/links/tampered-image-detection-and-localization/w5fy6lb0

(tampered vs. authentic), achieving 96.21% accuracy on CASIA v2.0 (JPEG-only subset). However, classification-only models cannot localize the tampered region — they answer "is it tampered?" but not "where?"

C. Segmentation-Based Localization

To achieve pixel-level localization, researchers adopted encoder-decoder architectures. The UNet [4], originally designed for biomedical image segmentation, proved effective for tamper localization. Using pretrained encoders (e.g., ResNet [5] trained on ImageNet [11]) significantly improves performance on small forensic datasets by providing rich, generalizable feature representations. Nested variants like UNet++ [16] further improve boundary delineation through dense skip connections.

D. Attention Mechanisms

Attention modules help the model focus on the most relevant features. CBAM (Convolutional Block Attention Module) [6] applies both channel-wise and spatial attention, teaching the model which feature maps and which spatial locations are most important. In our experiments, CBAM provided one of the most parameter-efficient improvements.

E. Multi-Stream and Frequency-Domain Methods

Recent state-of-the-art systems such as EMT-Net [10] combine multiple input streams (RGB, noise maps, frequency features) using Swin Transformers, achieving AUC of 0.987 on the NIST dataset. SRM (Spatial Rich Model) noise filters [9] extract high-pass noise residuals that reveal manipulation traces invisible in normal images. Bayar and Stamm [17] proposed constrained convolutional layers that learn manipulation-specific features, while ManTra-Net [18] introduced anomaly detection for forgery localization without requiring per-dataset fine-tuning. These techniques represent the current frontier, but require significantly more computational resources.

III. Dataset

A. CASIA v2.0

All experiments use the **CASIA v2.0** dataset [1], a widely-used benchmark for image forgery detection and localization. Table I summarizes its key properties.

TABLE I
CASIA v2.0 Dataset Summary

Property	Value
Total images	12,614
Authentic images	7,491 (59.4%)
Tampered images	5,123 (40.6%)
Copy-move forgeries	3,295
Splicing forgeries	1,828
Ground truth	Binary masks for tampered images
Input resolution	Resized to 384×384

B. Data Splitting

The dataset is split using stratified sampling (preserving the authentic/tampered ratio) with a fixed random seed of 42 for reproducibility:

TABLE II
Data Split Configuration

Split	Count	Purpose
Training	8,829 (70%)	Model training
Validation	1,892 (15%)	Early stopping & LR scheduling
Test	1,893 (15%)	Final reported metrics

C. Data Challenges

Several challenges make this dataset difficult:

- **Pixel-level class imbalance:** Most pixels in tampered images are still authentic. The tampered region is often very small.
- **Forgery type imbalance:** Copy-move forgeries (3,295) significantly outnumber splicing forgeries (1,828).
- **Compression artifacts:** JPEG artifacts can act as shortcuts, leading to data leakage if not handled carefully.

IV. Methodology

The final system consists of three main components: ELA preprocessing, a UNet segmentation model with a pretrained encoder, and a training pipeline with carefully chosen loss functions and hyperparameters.

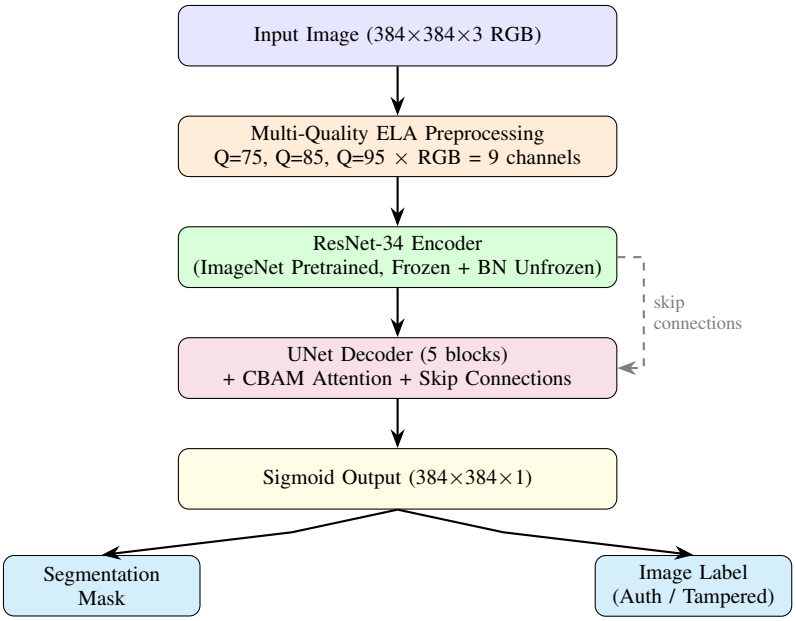

Fig. 1. Architecture of the proposed tamper detection and localization pipeline. The input RGB image is preprocessed with multi-quality ELA to produce a 9-channel forensic representation. The frozen pretrained ResNet-34 encoder extracts hierarchical features, which are passed through the UNet decoder with CBAM attention and skip connections. The sigmoid output produces a pixel-level probability map; image-level classification is derived from the maximum pixel probability.

A. Error Level Analysis (ELA) Preprocessing

ELA is a forensic preprocessing technique that reveals compression inconsistencies in images. It works as follows:

1) Load the original image as RGB.
2) Re-save it as a JPEG at a specific quality level Q.
3) Reload the recompressed image.
4) Compute the pixel-wise absolute difference between the original and the recompressed version.
5) Scale the brightness so the maximum difference maps to 255.

Authentic regions compress uniformly and show low ELA residuals. Tampered regions, which were saved at different compression levels or composited from different sources, show inconsistencies that appear as bright artifacts in the ELA map.

Multi-Quality RGB ELA (Best Variant): Instead of using a single quality level, the best configuration uses three quality levels — $Q = 75$, $Q = 85$, and $Q = 95$ — each producing an RGB ELA map. These three maps are stacked to form a 9-channel input tensor (3 qualities $\times$ 3 RGB channels $=$ 9 channels). Each quality level captures different types and strengths of compression artifacts:

- **Q=75 (aggressive)**: Produces large residuals; good at catching strong manipulations.
- **Q=85 (balanced)**: Medium residuals; a general-purpose level.
- **Q=95 (gentle)**: Small residuals; sensitive to subtle edits.

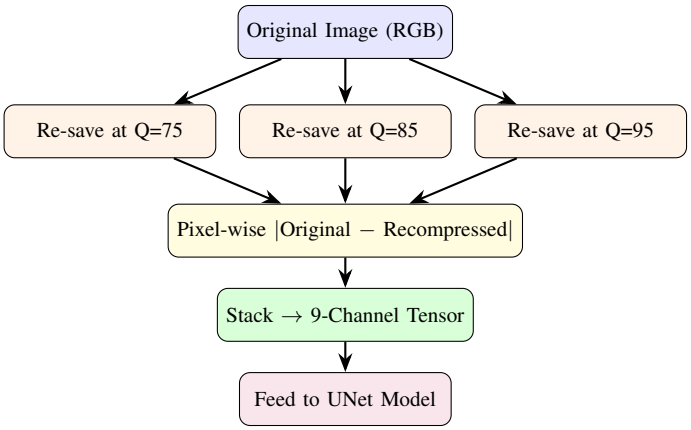

Fig. 2. Multi-Quality RGB ELA preprocessing pipeline. The original image is recompressed at three JPEG quality levels (Q=75, Q=85, Q=95). The absolute difference between the original and each recompressed version captures compression inconsistencies at different sensitivity levels. The three RGB difference maps are stacked into a 9-channel tensor used as input to the model.

B. Model Architecture

The model is a **UNet** [4] encoder-decoder segmentation network, implemented using the Segmentation Models PyTorch (SMP) library [7].

- **Encoder:** ResNet-34 [5] pretrained on ImageNet. All convolutional weights are *frozen*; only BatchNorm layers are set to training mode (unfrozen). This allows the normalization statistics to adapt from natural images to ELA forensic images, while keeping the learned feature hierarchies intact. The first convolutional layer is adapted to accept 9 input channels.
- **Decoder:** 5 upsampling blocks with channel dimensions (256, 128, 64, 32, 16) and skip connections at 4 resolution levels. Optional CBAM attention modules are placed after each decoder block.
- **Output:** A single-channel sigmoid activation producing a 384$\times$384 pixel probability map. A threshold of 0.5 converts this to a binary mask.
- **Image-level classification:** Derived from the segmentation output — if any pixel in the predicted mask exceeds the threshold, the image is classified as tampered.

TABLE III
MODEL PARAMETER SUMMARY

Component	Parameters
Total parameters	24.4M
Trainable parameters	3.17M (13%)
Encoder (frozen convolutions)	21.3M
Decoder (trainable)	∼3.17M
CBAM attention (optional)	+11,402 (+0.36%)

C. CBAM Attention Module

CBAM (Convolutional Block Attention Module) [6] is injected into all 5 UNet decoder blocks. It applies two sequential attention operations:

1) **Channel attention:** Learns *which feature channels* are most relevant. Uses a shared MLP (with bottleneck ratio 16) on both average-pooled and max-pooled features, combined with sigmoid gating.
2) **Spatial attention:** Learns *which spatial locations* to focus on. Concatenates channel-wise average and max pooling, processes with a 7$\times$7 convolution, and applies sigmoid.

CBAM adds only 11,402 parameters (0.36% of trainable parameters) but produced a +3.57 percentage point improvement in Pixel F1 — the highest improvement-per-parameter of any change tested.

D. Training Configuration

Table IV lists the training hyperparameters used across all experiments in the primary research track (vR.P).

TABLE IV
TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	Adam [14] (weight decay 10^{-5})
Learning rate	10^{-3}
Loss function	BCE + Dice [12]
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 3)
Early stopping	Patience 7 on validation loss
Batch size	16
Max epochs	25 (50 for extended runs)
Mixed precision	AMP [15] + TF32
Hardware	Kaggle T4 / P100 GPUs
Random seed	42

E. Evaluation Metrics

We use the following metrics to evaluate model performance, computed on **tampered images only** (since authentic images trivially score perfect masks):

- **Pixel F1:** The primary metric. Harmonic mean of pixel-level precision and recall. Measures how well the predicted mask overlaps with the ground truth mask.
- **IoU (Jaccard):** Intersection over Union between predicted and ground truth masks. A stricter metric than F1.
- **Pixel AUC:** Area under the ROC curve at the pixel level. Threshold-independent measure of discrimination quality.
- **Image Accuracy:** Binary classification accuracy — what fraction of images are correctly labeled as authentic or tampered.

An important methodological note: early experiments computed metrics across all images (both authentic and tampered), which *inflated* results because authentic images with all-zero ground truth masks always score perfect Dice/IoU/F1. From experiment vR.P.3 onward, all segmentation metrics are computed on tampered images only.

V. Research Journey

This project did not arrive at its final approach on the first attempt. The path included multiple failed approaches, each teaching lessons that directly shaped the next one. We document all four tracks here because the failures were as informative as the successes.

A. Track 1: Documentation-First (v0x) — Failed

The project began with an extensive literature review. Documentation was written before any code: architecture designs, dataset analyses, and training plans. Too many ideas from research papers were added simultaneously, leading to a complex system on paper that was never tested.

Lesson: Documentation without experimentation leads to untested assumptions. Start coding and validating early.

B. Track 2: Kaggle Notebook Reproduction (vK) — Failed

A promising Kaggle notebook for image tampering detection was found and adapted. However, an audit revealed **data leakage** — test images were leaking into training, inflating reported accuracy to $\sim$89.75%. After fixing the leakage, the model (training from scratch on $\sim$10K images with 31.6M parameters) achieved only Tampered F1 = 0.4101.

Lesson: Always audit data pipelines for leakage. Training from scratch on small datasets is insufficient.

C. Track 3: Research Paper Reproduction (vR) — Archived

The project pivoted to reproducing a published paper [3], achieving a best classification accuracy of 90.23%. However, this architecture was classification-only — it could not produce a localization mask.

Lesson: Verify that the base architecture supports all required tasks before investing in ablation studies.

D. Track 4: Pretrained Ablation Study (vR.P) — Final Approach

Armed with lessons from all prior failures, the final approach used:

- A **pretrained encoder** (ResNet-34 on ImageNet) to avoid the small-dataset problem.
- A **UNet segmentation model** for pixel-level localization.
- **ELA preprocessing** (from the ETASR track insight) to give the model forensic input.
- **Single-variable ablation:** exactly one change per experiment.
- **Weights & Biases** for experiment tracking and reproducibility.

This is the track that produced all final results reported below.

VI. Results

A. Best Model Performance

The best model (vR.P.19, Multi-Quality RGB ELA 9-channel input) achieves:

TABLE V
Best Model Results (vR.P.19)

Metric	Value
Pixel F1	**0.7965**
IoU (Jaccard)	0.6615
Pixel AUC	0.9665

B. Top 5 Runs

Table VI shows the five best-performing experiments.

TABLE VI
Top 5 Experiment Results by Pixel F1

Rank	Version	Pixel F1	Configuration
1	vR.P.19	**0.7965**	Multi-Q RGB ELA 9ch
2	vR.P.30.1	0.7762	Multi-Q ELA + CBAM, 50ep
3	vR.P.30.4	0.7745	Multi-Q ELA + CBAM + aug
4	vR.P.30.2	0.7721	Multi-Q ELA + CBAM + unfreeze
5	vR.P.30	0.7714	Multi-Q ELA + CBAM, 25ep

C. Key Experiment Results

Table VII summarizes the most important experiments from the ablation study. Each row represents exactly one change from the baseline. The full W&B experiment dashboard is available online.[3]

[3]https://api.wandb.ai/links/tampered-image-detection-and-localization/w5fy6lb0

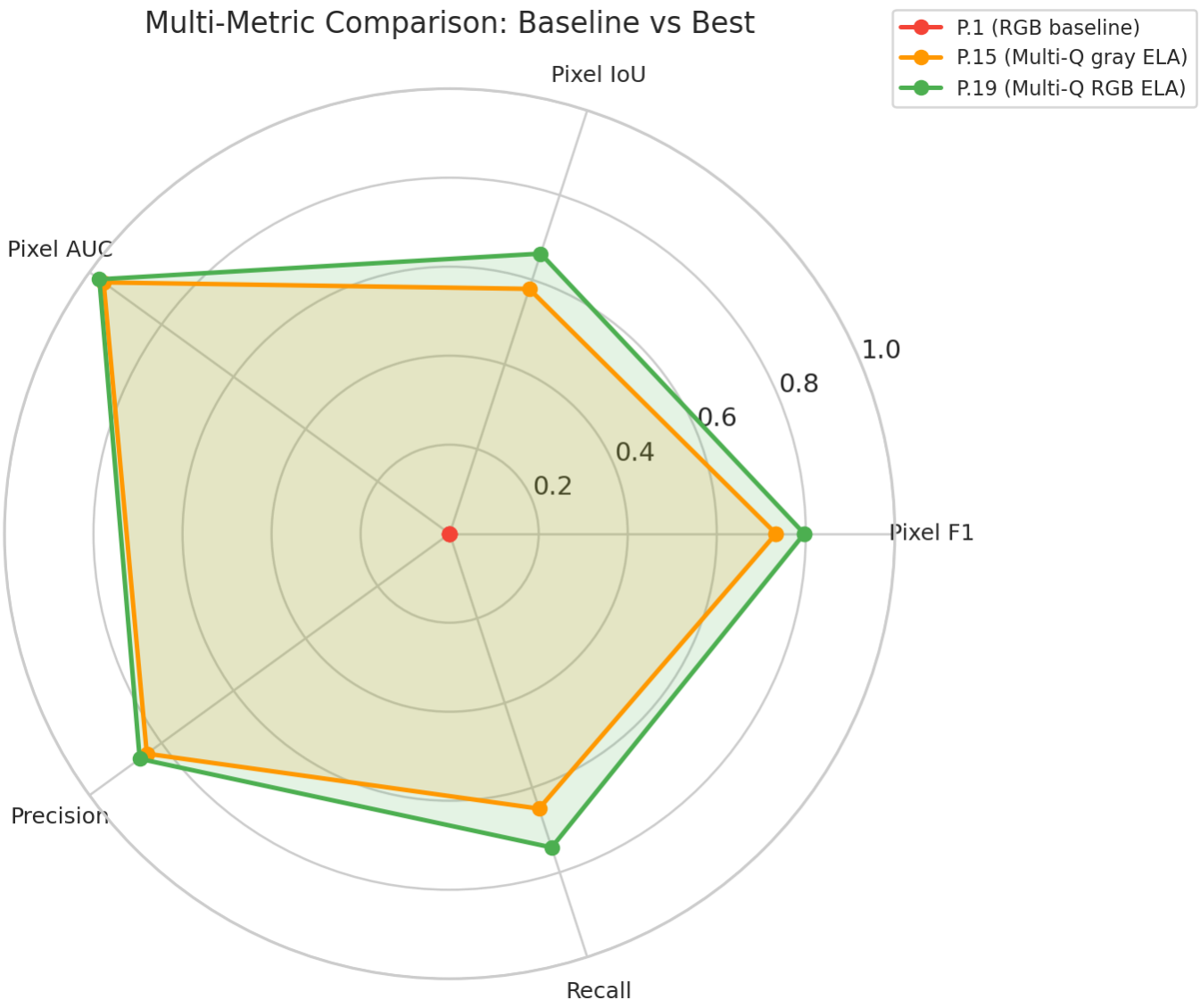

Fig. 3. Multi-metric radar comparison across three key configurations: RGB baseline (P.1), ELA baseline (P.3), and best model (vR.P.19). Each axis represents a different performance metric (Pixel F1, IoU, Pixel AUC, Image Accuracy). The area enclosed by vR.P.19 is substantially larger, confirming its dominance across all evaluation criteria.

TABLE VII
KEY EXPERIMENT RESULTS FROM THE ABLATION STUDY. EACH ROW IS A SINGLE-VARIABLE CHANGE. BOLD ROWS MARK THE MOST SIGNIFICANT FINDINGS.

Ver.	What Changed	Px F1	IoU	Verdict
P.1	RGB baseline (proper GT)	.4546	.2942	Baseline
P.3	**ELA input (replace RGB)**	**.6920**	**.5291**	**Strong +**
P.5	ResNet-50 encoder (RGB)	.5137	.3456	Positive
P.6	EfficientNet-B0 (RGB)	.5217	.3529	Positive
P.7	Extended training (50ep)	.7154	.5569	Positive
P.9	Focal + Dice loss	.6923	.5294	Neutral
P.10	CBAM attention	.7277	.5719	Positive
P.12	Data augmentation	.6968	.5347	Neutral
P.14b	Test-Time Aug (TTA)	.6388	.4693	Negative
P.15	**Multi-Q ELA (gray)**	**.7329**	**.5785**	**Positive**
P.16	DCT spatial map	.3209	.1911	Negative
P.17	ELA + DCT fusion	.7302	.5751	Positive
P.19	**Multi-Q RGB ELA 9ch**	**.7965**	**.6615**	**Best**
P.27	JPEG compression aug	.7523	.6029	Positive
P.28	Cosine annealing LR	.7601	.6132	Positive
P.30.1	Multi-Q + CBAM (50ep)	.7762	.6344	2nd Best

VII. ABLATION STUDY

The ablation study is the core contribution of this work. Each experiment changed *exactly one variable* from a baseline, enabling clear cause-and-effect analysis. This section organizes findings by category.

A. Impact Hierarchy

Table VIII ranks all modifications by their impact on Pixel F1 relative to the ELA baseline (P.3, F1=0.6920).

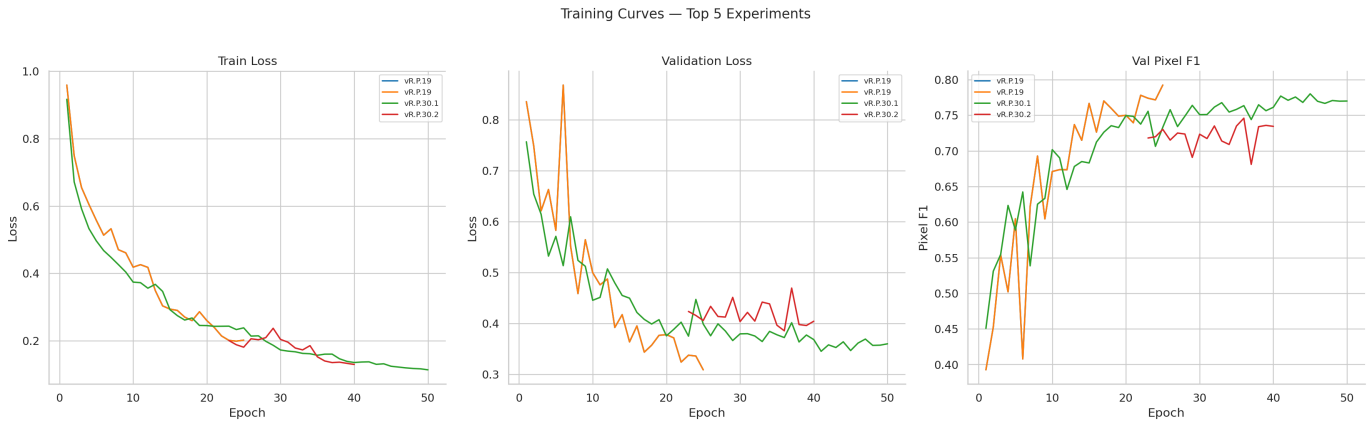

Fig. 4. Training loss and validation Pixel F1 curves for representative experiments. The curves illustrate convergence behavior: ELA-based models (P.3, P.15, P.19) converge to substantially higher F1 values than the RGB baseline (P.1), confirming that input representation is the dominant factor.

TABLE VIII
ABLATION IMPACT HIERARCHY (DELTA FROM ELA BASELINE P.3)

Rank	Category	Change	Δ F1 (pp)
1	Input	Multi-Q RGB ELA 9ch (P.19)	+10.45
2	Input	Multi-Q ELA gray (P.15)	+4.09
3	Input	ELA+DCT fusion (P.17)	+3.82
4	Architecture	CBAM attention (P.10)	+3.57
5	Training	Extended training (P.7)	+2.34
6	Input	RGB+ELA fusion (P.4)	+1.33
7	Training	Progressive unfreeze (P.8)	+0.65
8	Training	Augmentation (P.12)	+0.48
9	Loss	Focal+Dice (P.9)	+0.03
10	Evaluation	TTA (P.14b)	**−5.32**
11	Input	DCT-only (P.16)	**−37.11**

The hierarchy is clear and consistent:

Input representation > Attention mechanisms > Training configuration > Loss function

B. Input Representation (Dominant Factor)

This was the most impactful category across the entire study.

- **RGB to ELA (P.3):** +23.74pp Pixel F1. The single largest improvement in the study. False positive rate dropped from 22.6% to 2.7%.
- **Multi-Q ELA grayscale (P.15):** +4.09pp from P.3. Three grayscale ELA maps at Q=75/85/95 capture complementary forensic signals.
- **Multi-Q RGB ELA 9-channel (P.19):** +10.45pp from P.3. Keeping full RGB color information in each ELA map (rather than converting to grayscale) was worth an additional +6.36pp over the grayscale variant.
- **DCT-only (P.16):** −37.11pp. Catastrophic failure. Block-level frequency features alone are completely insufficient.

C. Encoder Architecture

- **CBAM attention (P.10):** +3.57pp from P.3. Best architectural modification. Achieved the lowest false positive rate (2.0%).
- **ResNet-50 (P.5):** F1=0.5137 on RGB (2.86× more parameters, still below ELA on ResNet-34).
- **EfficientNet-B0 (P.6):** F1=0.5217 on RGB (smallest model, but still far below ELA results).

This demonstrates that **ELA input on ResNet-34** outperforms larger encoders on raw RGB — input preprocessing matters more than encoder capacity.

D. Training Strategies

- **Extended training (P.7):** +2.34pp. The 25-epoch default was premature; the best epoch was at epoch 36.
- **Data augmentation (P.12):** +0.48pp Pixel F1 but achieved the best image-level accuracy (88.48%).
- **Focal+Dice loss (P.9):** +0.03pp (essentially neutral). Focal Loss [13] concentrates on hard examples, but the Dice component already handles class imbalance.
- **TTA (P.14b):** −5.32pp. Averaging across flipped views smooths sharp mask boundaries, destroying localization precision.

E. Diminishing Returns

After Pixel F1 ∼0.78, combining improvements shows diminishing returns. For example, vR.P.30.1 (Multi-Q ELA + CBAM + 50 epochs) = 0.7762, which did *not* surpass P.19's 0.7965. This suggests the model approaches the ceiling of what the dataset and architecture can support.

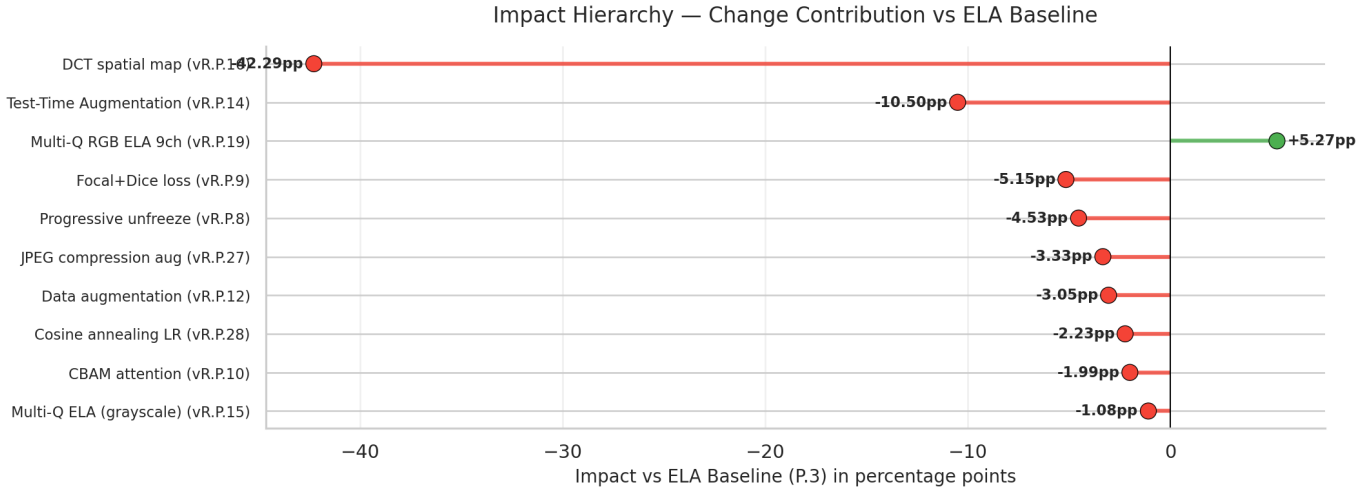

Fig. 5. Visual impact hierarchy: Pixel F1 delta (percentage points) for each experimental modification relative to the ELA baseline (P.3, F1=0.6920). Positive bars indicate improvement; negative bars indicate degradation. Multi-Quality RGB ELA (P.19) and grayscale ELA (P.15) yield the largest gains; DCT-only (P.16) and TTA (P.14b) hurt performance.

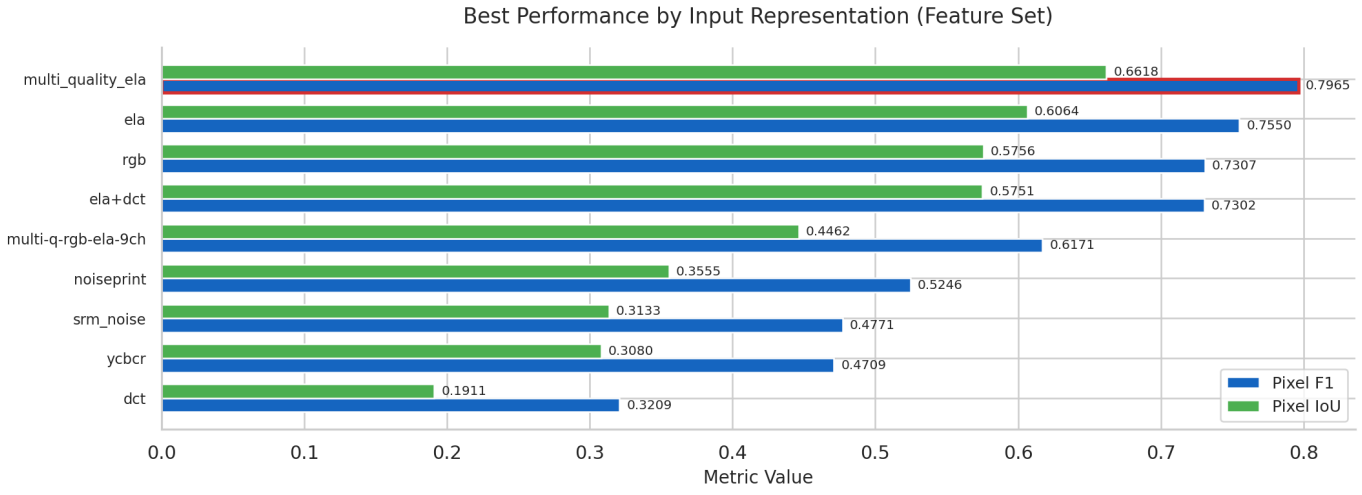

Fig. 6. Best Pixel F1 and IoU achieved by each input representation type. Multi-Quality RGB ELA (9 channels) achieves the highest scores across both metrics, while DCT-only fails catastrophically. The chart confirms that input preprocessing is the single most impactful design decision in tamper localization.

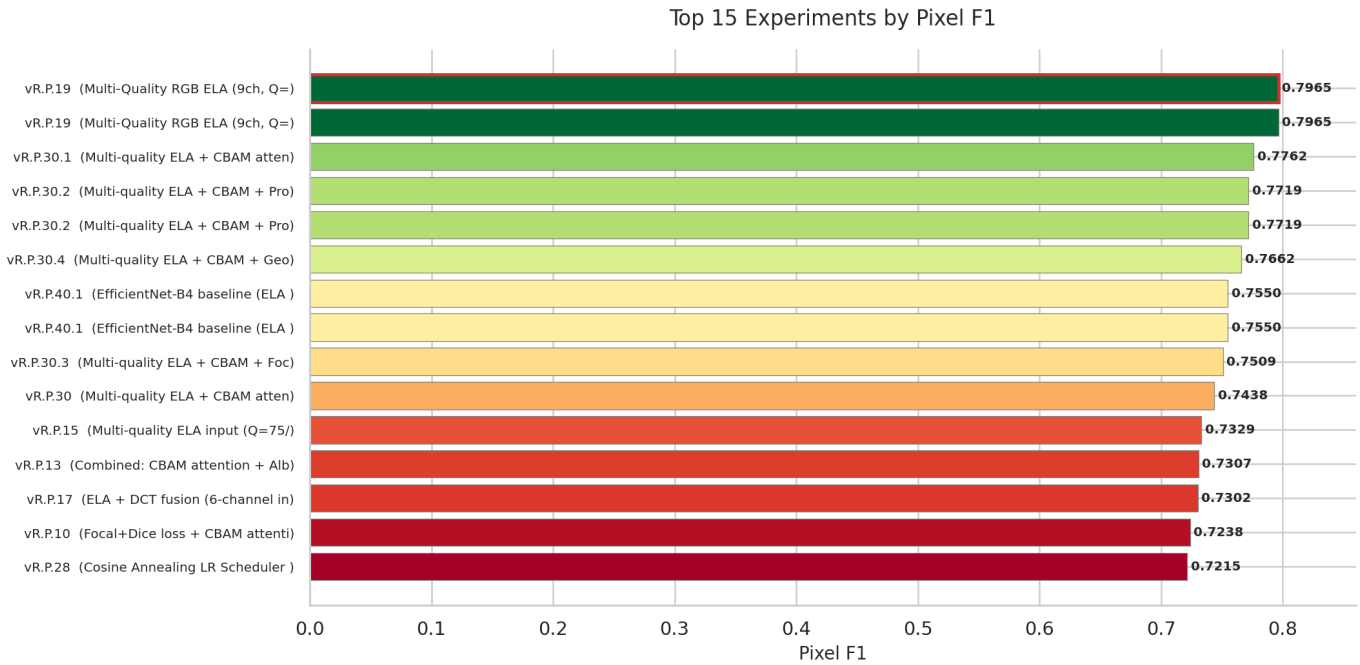

Fig. 7. Top 15 experiments ranked by Pixel F1 score. vR.P.19 (Multi-Quality RGB ELA, 9-channel) leads the leaderboard. The cluster of vR.P.30.x combination experiments (CBAM + extended training) fall below P.19, illustrating diminishing returns when multiple improvements are combined after a strong input representation baseline.

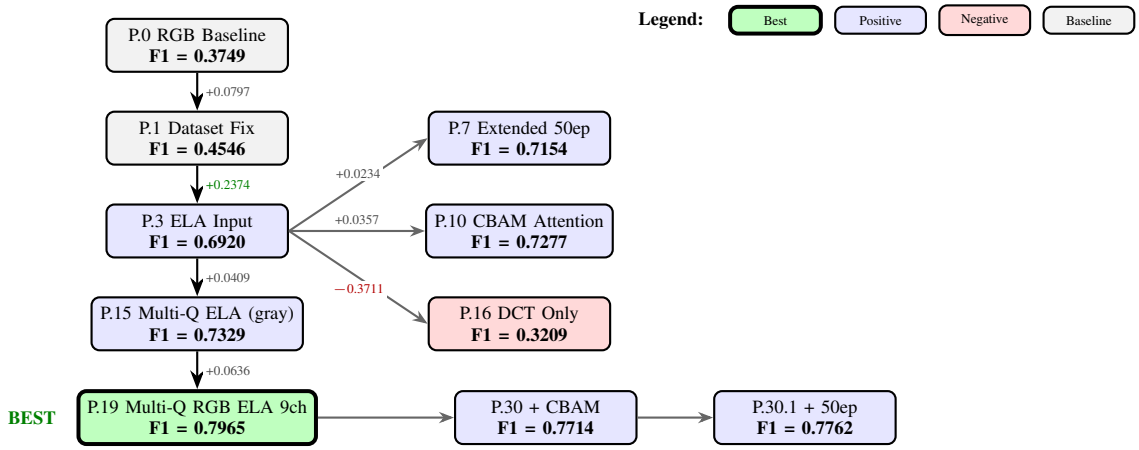

Fig. 8. Experiment lineage diagram for the vR.P research track. The main trunk (left) shows the primary evolution path from RGB baseline (P.0) to the best model (P.19, Pixel F1 = 0.7965). Branches show single-variable explorations from the ELA baseline (P.3). Delta annotations on arrows indicate the Pixel F1 change. The largest single improvement (+0.2374) was switching from RGB to ELA input. Green = best result; blue = positive; red = negative; gray = baseline.

VIII. Error Analysis

A. False Positive and False Negative Rates

Table IX shows how false positive (FP) and false negative (FN) rates evolved across key configurations.

TABLE IX
False Positive and False Negative Rates

Configuration	FP Rate	FN Rate
RGB baseline (P.1)	22.6%	40.4%
ELA baseline (P.3)	2.7%	28.6%
CBAM attention (P.10)	2.0%	28.3%
Data augmentation (P.12)	2.6%	24.6%
TTA (P.14b)	1.2%	29.3%

ELA reduces the false positive rate dramatically (22.6% $\rightarrow$ 2.7%). CBAM further reduces it to 2.0%. The remaining errors are primarily false negatives — the model misses some tampered regions, particularly:

- Subtle copy-move forgeries where source and target textures are very similar.
- Very small tampered regions (<2% of image area).
- Images that have been re-saved multiple times, destroying the ELA signal.

B. Best Model Detailed Metrics

Table X presents the full metric breakdown for the best model (vR.P.19), showing the precision-recall trade-off.

TABLE X
Detailed Metrics for the Best Model (vR.P.19)

Metric	Value
Pixel F1	0.7965
Pixel IoU	0.6618
Pixel AUC	0.9726
Pixel Precision	0.8122
Pixel Recall	0.7829
Image Accuracy	90.39%
Image Macro F1	0.8961
Image ROC-AUC	0.9726

The model achieves a strong balance of precision (0.8122) and recall (0.7829) — it accurately identifies tampered regions while successfully capturing the majority of tampered pixels.

IX. Reproducibility and Experiment Tracking

Reproducibility was a core design principle throughout this project.

A. Fixed Experimental Constants

Every experiment in the vR.P track shares the same dataset, random seed (42), data split (70/15/15 stratified), input resolution (384×384), decoder architecture (UNet with skip connections), batch size (16), and framework (PyTorch + SMP). Only one variable is changed per experiment.

B. Weights & Biases Integration

All experiments are logged to a centralized W&B project, tracking:

- **Per-epoch:** training loss, validation loss, validation Pixel F1, validation IoU, learning rate.
- **Final test:** all segmentation and classification metrics.
- **Media:** prediction visualization examples.

A centralized leaderboard notebook automatically queries the W&B API to generate up-to-date comparison tables and ablation charts.

C. Reproducibility Verification

Two experiments (vR.P.3 and vR.P.10) were run twice independently on Kaggle T4 GPUs. Both runs produced identical metrics, confirming the reproducibility of results with seed 42.

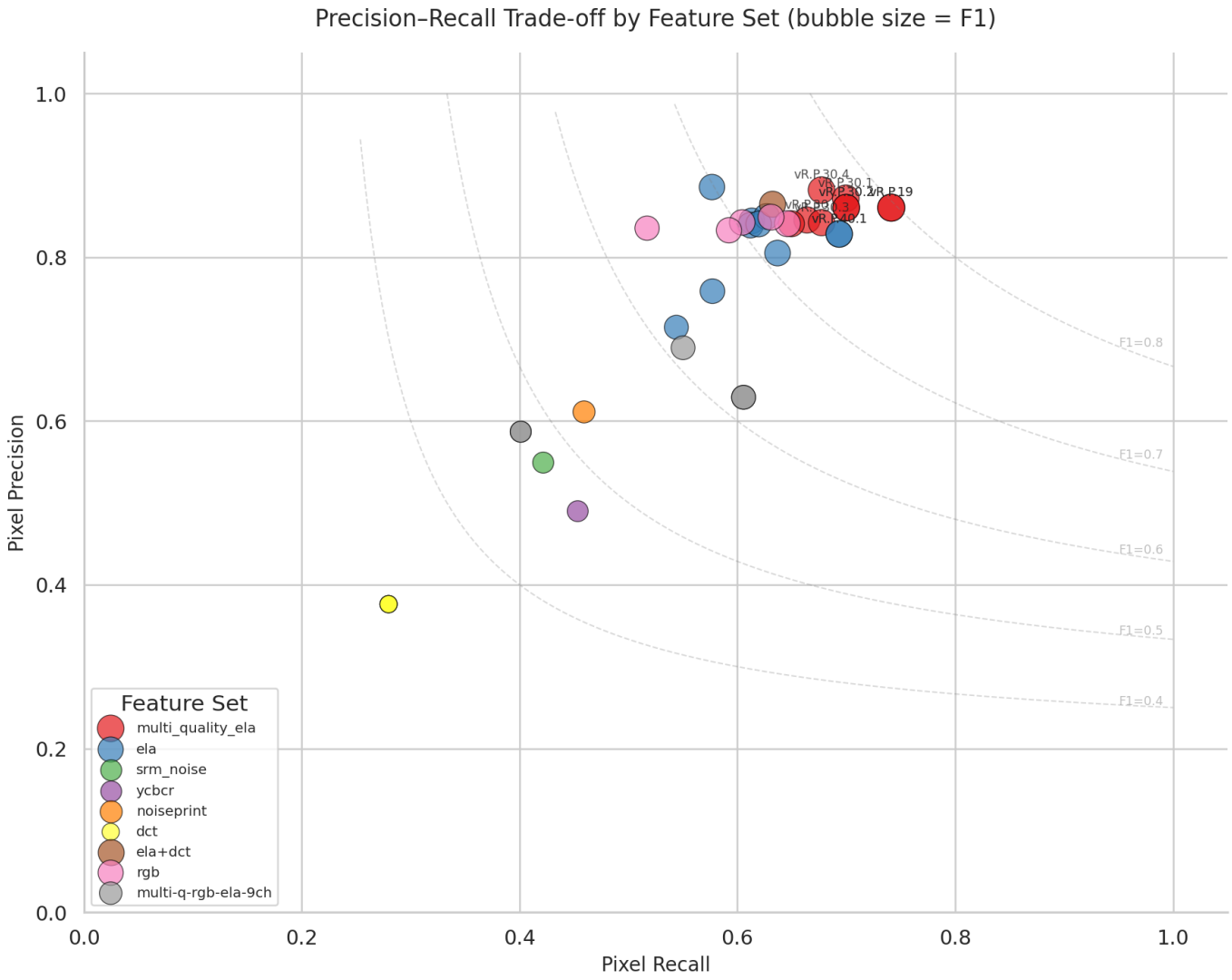

Fig. 9. Precision vs. recall scatter by feature set. ELA-based inputs (ELA, Multi-Q ELA Gray, Multi-Q RGB ELA) occupy the upper-right quadrant, indicating both high precision and high recall. The RGB baseline and DCT-only configurations are in the lower-left, confirming that input representation is the key determinant of the precision-recall balance.

X. Key Learnings

The following lessons emerged from the full experimental journey:

1) **Data leakage destroys credibility.** The Kaggle baseline appeared to work well but had inflated results due to test data leaking into training. Always audit data pipelines.
2) **Pretrained models are essential for small datasets.** Training from scratch on $\sim$10K images learns only low-level features. Pretrained encoders provide rich, generalizable representations.
3) **Input representation matters more than architecture.** ELA preprocessing (+23.74pp) and multi-quality ELA (+10.45pp more) produced far larger gains than any architectural change.
4) **Change one thing at a time.** Early attempts that added multiple improvements simultaneously led to confusion and failure. Single-variable ablation is the only way to understand what actually works.
5) **Negative results are valuable.** Knowing that DCT alone fails (-37.11pp) or that TTA hurts localization (-5.32pp) prevents wasted effort in future work.
6) **Verify the base approach supports all tasks.** An entire research track (vR.x.x) was discontinued because the architecture could only classify, not localize.
7) **Track experiments formally.** Weights & Biases [19] made it possible to compare 60+ experiments systematically with reproducible metrics.

XI. Limitations

1) **Single dataset:** All results are on CASIA v2.0 only. Cross-dataset generalization (Columbia, Coverage, NIST) is untested.
2) **Fixed resolution:** 384×384 input may lose fine details. Higher resolution (512×512+) may improve boundary precision.
3) **Binary segmentation only:** The model treats all tampering as one class. It does not distinguish between splicing and copy-move.
4) **No robustness testing:** Behavior against post-processing attacks (JPEG recompression, blur, noise addition) is not evaluated.
5) **No SRM noise features:** Spatial Rich Model noise extraction — used by state-of-the-art systems — is not yet implemented.
6) **Diminishing returns:** After Pixel F1 ∼0.80, the current architecture likely needs different techniques (transformers, larger datasets) to push further.

XII. Future Work

- **Cross-dataset evaluation:** Test on Columbia, Coverage, CoMoFoD, and NIST datasets to verify generalization.
- **Transformer-based encoders:** Swin Transformer or ConvNeXt may capture long-range manipulation patterns better than ResNet.
- **SRM noise filters:** Spatial Rich Model high-pass filters for manipulation trace extraction in the noise domain.
- **Higher resolution:** Test at 512×512 or 640×640 to capture finer boundary details.
- **Ensemble methods:** Combine predictions from multiple ELA variants, CBAM, and augmentation configurations.
- **Multi-class segmentation:** Distinguish splicing from copy-move at the pixel level.

XIII. Conclusion

This report presented a systematic ablation study for image tamper detection and localization on the CASIA v2.0 dataset. Through 60+ controlled experiments across four research tracks, we demonstrated that:

1) **Input representation is the dominant factor.** Switching from raw RGB to Multi-Quality RGB ELA (9-channel) produced a +34.19 percentage point improvement in Pixel F1 — far exceeding any architectural or training change.
2) **The best model achieves Pixel F1 = 0.7965, IoU = 0.6615, and Pixel AUC = 0.9665**, using a UNet with a frozen pretrained ResNet-34 encoder and 9-channel multi-quality ELA input.
3) **Single-variable ablation discipline** transforms ad-hoc experimentation into systematic scientific investigation, enabling clear cause-and-effect understanding.
4) **Failures are as informative as successes.** Each failed research track contributed lessons that directly improved the next approach.

The impact hierarchy established by this study — **Input representation** > **Attention mechanisms** > **Training configuration** > **Loss function** — provides a clear roadmap for future work: invest in better input representations before trying more complex architectures.

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